

BABU BANARASI DAS

UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

School Student Management System Using MongoDB

Submitted to:

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Department: BCA (DS & AI)

Date: 10th October

Submitted by:

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1. CREATE DATABASE

MongoDB Query:

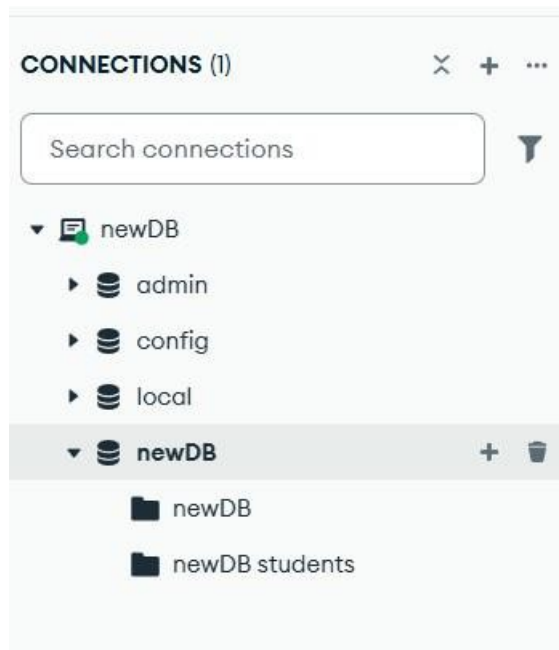
use newDB

```
> _MONGOSH  
  
> use("newDB")  
< switched to db newDB  
  
> db.newDB.insertMany([  
  {roll:1, name:"Drishti", age:19,marks:88,city:"Lucknow"},
```

2. CREATE COLLECTION

MongoDB Query:

db.create Collection(" newDB students")



3. INSERT 50 STUDENTS

MongoDB Query:

```
db.newDB. insertMany([ ])
```

```
{roll:1, name:"Drishti", age:19,marks:88,city:"Lucknow"},  
{roll:2, name:"Ayushi", age:18,marks:76,city:"Jaunpur"},  
{roll:3, name:"Vaishali", age:19,marks:92,city:"Kanpur"},  
{roll:4, name:"Annu", age:18,marks:69,city:"Ayodhya"},  
{roll:5, name:"Vansh", age:19,marks:81,city:"Varanasi"},  
{roll:6, name:"Amrita", age:20,marks:85,city:"Lucknow"},  
{roll:7, name:"Naveen", age:19,marks:73,city:"Prayagraj"}, {roll:8,  
name:"Pratik", age:19,marks:90,city:"Lucknow"},  
{roll:9, name:"Sangita", age:20,marks:78,city:"Gorakhpur"},  
{roll:10, name:"Vivek", age:18,marks:65,city:"Kanpur"},  
{roll:11, name:"Tarkeshwar", age:20,marks:71,city:"Jaunpur"},  
{roll:12, name:"Shreya", age:19,marks:95,city:"Lucknow"},  
{roll:13, name:"Akshita", age:18,marks:84,city:"Ayodhya"},  
{roll:14, name:"Shruti", age:19,marks:87,city:"Prayagraj"},  
{roll:15, name:"Kartik", age:20,marks:74,city:"Varanasi"},  
{roll:16, name:"Sejal", age:18,marks:89,city:"Lucknow"},  
{roll:17, name:"Astha", age:19,marks:82,city:"Kanpur"},  
{roll:18, name:"Jai", age:18,marks:58,city:"Gonda"},  
{roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},  
{roll:20, name:"Omkareshwar", age:19,marks:63,city:"Bareilly"},  
{roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"}, {roll:22,  
name:"Saumya", age:19,marks:86,city:"Jaunpur"}, {roll:23,  
name:"Vidya", age:20,marks:72,city:"Kanpur"},  
{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},
```

{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"}, {roll:26, name: "Anushka",
age:20,marks:94,city:"Lucknow"},
{roll:27, name: "Ananya", age:19,marks:88,city:"Varanasi"},
{roll:28, name: "Bhaskar", age:18,marks:61,city:"Gorakhpur"},
{roll:29, name: "Rohan", age:19,marks:75,city:"Kanpur"},
{roll:30, name: "Nishant", age:20,marks:79,city:"Lucknow"},
{roll:31, name: "Vineet", age:18,marks:68,city:"Jaunpur"},
{roll:32, name: "Anika", age:19,marks:90,city:"Lucknow"},
{roll:33, name: "Jai", age:20,marks:55,city:"Ayodhya"},
{roll:34, name: "Lakshya", age:18,marks:83,city:"Kanpur"},
{roll:35, name: "Krishna", age:19,marks:96,city:"Prayagraj"},
{roll:36, name: "Rumana", age:20,marks:70,city:"Lucknow"},
{roll:37, name: "Priyanshu", age:18,marks:81,city:"Varanasi"},
{roll:38, name: "Aditya", age:19,marks:66,city:"Gorakhpur"},
{roll:39, name: "Dhruv", age:20,marks:93,city:"Lucknow"},
{roll:40, name: "Nivedita", age:19,marks:85,city:"Kanpur"},
{roll:41, name: "Meenu", age:18,marks:74,city:"Jaunpur"},
{roll:42, name: "Akanksha", age:20,marks:89,city:"Lucknow"},
{roll:43, name: "Sarika",age:19,marks:62,city:"Ayodhya"},
{roll:44, name: "Anchal",age:18,marks:78,city:"Prayagraj"},
{roll:45, name: "Umra", age:20,marks:84,city:"Lucknow"},
{roll:46, name: "Nashra", age:19,marks:73,city:"Kanpur"},
{roll:47, name: "Aman", age:18,marks:59,city:"Gonda"},
{roll:48, name: "Priya", age:20,marks:87,city:"Lucknow"},
{roll:49, name: "Rahul", age:19,marks:64,city:"Bareilly"},
{roll:50, name: "Neha", age:18,marks:91,city:"Varanasi"}

> _MONGOSH

> use("newDB")

< switched to db newDB

> db.newDB.insertMany([

```
{roll:1, name:"Drishti", age:19,marks:88,city:"Lucknow"},
{roll:2, name:"Ayushi", age:18,marks:76,city:"Jaunpur"},
{roll:3, name:"Vaishali", age:19,marks:92,city:"Kanpur"},
{roll:4, name:"Annu", age:18,marks:69,city:"Ayodhya"},
{roll:5, name:"Vansh", age:19,marks:81,city:"Varanasi"},
{roll:6, name:"Amrita", age:20,marks:85,city:"Lucknow"},
{roll:7, name:"Naveen", age:19,marks:73,city:"Prayagraj"},
{roll:8, name:"Pratik", age:19,marks:90,city:"Lucknow"},
{roll:9, name:"Sangita", age:20,marks:78,city:"Gorakhpur"},
{roll:10, name:"Vivek", age:18,marks:65,city:"Kanpur"},
{roll:11, name:"Tarkeshwar", age:20,marks:71,city:"Jaunpur"},
{roll:12, name:"Shreya", age:19,marks:95,city:"Lucknow"},
{roll:13, name:"Akshita", age:18,marks:84,city:"Ayodhya"},
{roll:14, name:"Shruti", age:19,marks:87,city:"Prayagraj"},
{roll:15, name:"Kartik", age:20,marks:74,city:"Varanasi"},
{roll:16, name:"Sejal", age:18,marks:89,city:"Lucknow"},
{roll:17, name:"Astha", age:19,marks:82,city:"Kanpur"},
{roll:18, name:"Jai", age:18,marks:58,city:"Gonda"},
{roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},
{roll:20, name:"Omkareshwar", age:19,marks:63,city:"Bareilly"},
{roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"},
{roll:22, name:"Saumya", age:19,marks:86,city:"Jaunpur"},
{roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},
{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},
{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"},
{roll:26, name: "Anushka", age:20,marks:84,city:"Lucknow"}])
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

db.newDB.find ()

```
db["newDB"].find()
{
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Drishti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af5'),
  roll: 2,
  name: 'Ayushi',
  age: 20,
  marks: 76,
  city: 'Jaunpur'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af6'),
  roll: 3,
  name: 'Vaishali',
  age: 17,
  marks: 92,
  city: 'Kanpur'
}
```

5. EQUAL TO — \$eq

MongoDB Query:

db.newDB.find ({marks: {\$eq: 88}})

>_MONGOSH

```
> db.newDB.find ({marks: {$eq: 88}})
```

```
< {
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Drishti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2b0e'),
  roll: 27,
  name: 'Ananya',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
{
  _id: ObjectId('6ab25b664d329654826b87fd'),
  roll: 1,
  name: 'Jagriti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
```

6. GREATER THAN — \$gt

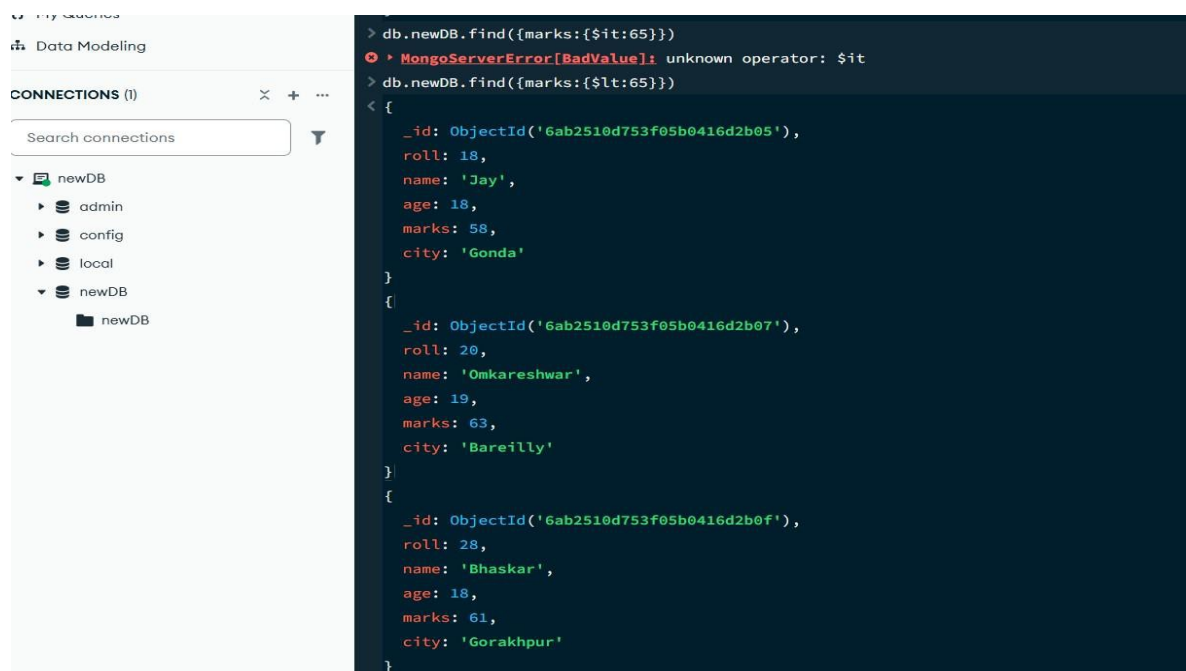
MongoDB Query:

```
db.newDB.find({marks: {$gt: 88}})
```

```
> db.newDB.find({marks:{$gt:88}})
< [
  {
    _id: ObjectId('6ab2510d753f05b0416d2afb'),
    roll: 3,
    name: 'Vaishali',
    age: 17,
    marks: 92,
    city: 'Kanpur'
  },
  {
    _id: ObjectId('6ab2510d753f05b0416d2afb'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  },
  {
    _id: ObjectId('6ab2510d753f05b0416d2aff'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
]
```

7. LESS THAN — \$lt

MongoDB Query: `db.newDB.find ({marks: {$lt: 65}})`



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'newDB' connection. The main panel displays the query `db.newDB.find({marks:{$lt:65}})` and its results. The results are an array of three documents:

```
> db.newDB.find({marks:{$lt:65}})
MongoServerError[BadValue]: unknown operator: $lt
> db.newDB.find({marks:{$lt:65}})
< [
  {
    _id: ObjectId('6ab2510d753f05b0416d2b05'),
    roll: 18,
    name: 'Jay',
    age: 18,
    marks: 58,
    city: 'Gonda'
  },
  {
    _id: ObjectId('6ab2510d753f05b0416d2b07'),
    roll: 20,
    name: 'Omkareshwar',
    age: 19,
    marks: 63,
    city: 'Bareilly'
  },
  {
    _id: ObjectId('6ab2510d753f05b0416d2b0f'),
    roll: 28,
    name: 'Bhaskar',
    age: 18,
    marks: 61,
    city: 'Gorakhpur'
  }
]
```


8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query: *db.newDB. find({marks: {\$gte: 90}})*

```
> _MONGOSH
}
> db.newDB.find({marks:{$gte:90}})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af6'),
  roll: 3,
  name: 'Vaishali',
  age: 17,
  marks: 92,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2afb'),
  roll: 8,
  name: 'Pratik',
  age: 19,
  marks: 90,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2aff'),
  roll: 12,
  name: 'Shreya',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2b06'),
  roll: 19,
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query: *db. students. find({marks: {\$lte: 70}})*

```
> db.newDB.find({marks:{$lte:70}})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af7'),
  roll: 4,
  name: 'Annu',
  age: 18,
  marks: 70,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2afd'),
  roll: 10,
  name: 'Vivek',
  age: 18,
  marks: 65,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2b05'),
  roll: 18,
  name: 'Jay',
  age: 18,
  marks: 58,
  city: 'Gonda'
}
```

10. UPDATE OPERATION

MongoDB Query:

```
db.newDB.updateOne(
  {roll:5},
  {$set:{marks:95}})
```

```
Welcome mongosh: newDB +
>_MONGOSH
> db.newDB.db.updateOne(
  {roll:5},
  {$set:{marks:95}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
> db.newDB.find({roll:5})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af8'),
  roll: 5,
  name: 'Vansh',
  age: 19,
  marks: 81,
  city: 'Varanasi'
}
```

11. DELETE ONE OPERATION MongoDB

Query:

```
db.newDB.deleteOne({roll:10})
```

```
> db.newDB.deleteOne({roll:10})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.newDB.find({roll:10})
<
```

12. DELETE MANY OPERATION

MongoDB Query: `db.newDB.deleteMany({ marks: {$lt:60}})`

```
> db.newDB.deleteMany({
  marks: {$lt:60}
})
< {
  acknowledged: true,
  deletedCount: 6
}
```

13. AND OPERATION — \$and

MongoDB Query: `db.newDB.find({
$and:[
 {marks:{$gt:80}},
 {age:19}
]})`

```
> db.newDB.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Drishti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af8'),
  roll: 5,
  name: 'Vansh',
  age: 19,
  marks: 81,
  city: 'Varanasi'
}
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.newDB.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]})
```

```
> db.newDB.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Jagriti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af6'),
  roll: 3,
  name: 'Vaishali',
  age: 17,
  marks: 92,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af9'),
  roll: 6,
  name: 'Amrita',
  age: 20,
  marks: 85,
  city: 'Lucknow'
```

15. SORT OPERATION

A.MongoDB Query: `db.newDB. find (). sort({marks : 1})`

```

> db.newDB. find (). sort ({marks : 1})
< {
  _id: ObjectId('6ab2510d753f05b0416d2b0f'),
  roll: 28,
  name: 'Bhaskar',
  age: 18,
  marks: 61,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6ab25d9eaa2d93a1b9ab099c'),
  roll: 28,
  name: 'Bhaskar',
  age: 18,
  marks: 61,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6ab25b664d329654826b8818'),
  roll: 28,
  name: 'Bhaskar',
  age: 18,
  marks: 61,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6ab25b664d329654826b8827'),
  roll: 42
}

```

B. MongoDB Query: `db.newDB. find (). sort ({marks: -1})`

```

> db.newDB. find (). sort ({marks: -1})
< {
  _id: ObjectId('6ab25d9eaa2d93a1b9ab09a3'),
  roll: 35,
  name: 'Krishna',
  age: 19,
  marks: 96,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2b16'),
  roll: 35,
  name: 'Krishna',
  age: 19,
  marks: 96,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6ab25b664d329654826b881f'),
  roll: 35,
  name: 'Krishna',
  age: 19,
  marks: 96,
  city: 'Prayagraj'
}

```


16. LIMIT OPERATION

MongoDB Query: *db.newDB.find().limit(5)*

```
> db.newDB.find().limit(5)
< {
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Drishti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af5'),
  roll: 2,
  name: 'Ayushi',
  age: 20,
  marks: 76,
  city: 'Jaunpur'
}
```

17. SEARCH BY CITY

MongoDB Query:

db.newDB.find({city:"Lucknow"})

```
> db.newDB.find({city:"Lucknow"})
< {
  _id: ObjectId('6ab2510d753f05b0416d2af4'),
  roll: 1,
  name: 'Drishti',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab2510d753f05b0416d2af9'),
  roll: 6,
  name: 'Amrita',
  age: 20,
  marks: 85,
  city: 'Lucknow'
}
```

18. COUNT DOCUMENTS

MongoDB Query: *db. newDB.*

countDocuments ()

```
> db. newDB. countDocuments ()
< 140
newDB >
```